

Fan Xiao

Tel: +49 15231698999, Email: Fan.Xiao@med.uni-muenchen.de, Website: <https://fansomx.github.io>

Education

Ludwig-Maximilians-Universität München, Munich, Germany PhD candidate in Medical Physics <i>Research focus:</i> AI-accelerated Dose Calculation, Monte Carlo Simulation, MRI Segmentation/Registration	2023.09 – Now
Southern Medical University, Guangzhou, China MS in Biomedical Engineering, GPA 3.6/4.0	2020.09 – 2023.06
Wuhan University of Technology, Wuhan, China BS in Communication Engineering	2016.09 – 2020.06

Research Experience

Multi-model study of fast VMAT segment dose calculation First Author - Designed a GPU-accelerated processing pipeline to perform ray-tracing segment projection and Beam's Eye View transformations - Developed a unified benchmarking framework comparing CNN-Mamba, Transformer, CNN-ConvLSTM, and Unet for dose calculation modeling - Achieving dose inference (<40 ms/segment) with low memory footprint (2.9 GB), validated across multiple GPU generations - Published in <i>Physics in Medicine & Biology</i> (2026), DOI: 10.1088/1361-6560/ac6413	2025.07–2026.04
DoseRAD2026 MICCAI Challenge Dataset First Author - Generate and co-organize a public paired CT and MRI dataset with beam-level photon and proton Monte Carlo dose distributions for developing and evaluating advanced dose calculation methods - Pre-processing: deformable image registration, air-cavity correction, matRad photon/proton planning - Ground-truth photon (6 MV) and proton dose distributions were computed using open-source Geant4 Monte Carlo simulations, yielding 40,500 photon beams and 81,000 proton beamlets - Submitted in <i>Medical Physics</i> (2026), DOI: https://arxiv.org/abs/2604.12778	2025.07–2026.04
Deep learning-based synthetic-CT-free photon dose calculation First Author - Proposed a UNet-LSTM framework to predict photon dose directly from MRI data - Explored the domain shift challenge (lack of electron density information in MRI) by integrating CT-MRI deformable registration with air cavity segmentation - Achieved 12 ms per photon beamlet inference time with a gamma passing rate (2%/2mm) exceeding 99.50% compared to Monte Carlo ground truth - Published in <i>Medical Physics</i> (2025), DOI: 10.1002/mp.70106	2024.10 – 2025.06
Proton Range Verification via LSTM-based Prompt Gamma Prediction First Author - Architected an LSTM model by integrating relative stopping power maps and proton dose to predict prompt gamma emissions across wide proton energies (125–210 MeV) - Achieved sub-millimeter accuracy with a mean absolute range shift deviation of 0.15 mm against Monte Carlo ground truth	2024.01 – 2024.09

- Reduced prediction time to <130 ms per pencil beam, facilitating sub-second feedback for adaptive proton therapy
- Published in *Physics in Medicine & Biology* (2024), DOI: [10.1088/1361-6560/ad8e2a](https://doi.org/10.1088/1361-6560/ad8e2a)

Simultaneous dose and fluence prediction for head and neck IMRT | Co-Author

2022.07 – 2023.05

- Developed a shared-encoder 3D U-Net for simultaneous dose & fluence prediction in H&N IMRT
- Achieved fluence MAE $0.53\% \pm 0.13\%$, SSIM 0.96 ± 0.02 , dose error <1 Gy for target/OAR structures
- Enabled automated IMRT planning pipeline by using predicted dose as optimization objective and fluence as warm-start initialization
- Published in *Radiation Oncology* (2023), DOI: [10.1186/s13014-023-02287-4](https://doi.org/10.1186/s13014-023-02287-4)

Technical Skills

- Programming languages: Python, C++ (Geant4), Bash
- Deep Learning: Pytorch, LSTM, Transformer, Mamba, Unet/nnUnet
- Tools: Git, Docker, Slurm, Cupy, Plastimatch, Pydicom, matRad
- TPS planning experience: RayStation, Varian Eclipse

Additional Info and Awards

- Conference experience:
ESTRO 2026 Poster (Stockholm), AAPM 2025 Oral (Washington, DC),
ICCR 2024 Oral (Lyon), ECMP 2024 Oral (Munich)
 - Awards:
Chinese National Scholarship (2022) – Top 2%
First-class Academic Scholarship (2022, 2023) – Top 20%
 - Languages:
Chinese (Native), English (Professional Working Proficiency)
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